# Probability · 4-1 & 4-2 · One-Page Sheet



## 4-1 · KEY TERMS (DEFINE AS SENTENCES)

- **Probability experiment**: a chance process with well-defined results (outcomes).
- **Outcome**: the result of a single trial.
- **Sample space S**: the set of all possible outcomes; size n(S).
- **Event**: a set of outcomes (a subset of S).
- **Simple** event = 1 outcome · **compound** = 2 or more.

النتيجة لمحاولة واحدة، وفضاء العينة كل النتائج، والحدث مجموعة جزئية منه.

## 4-1 · SAMPLE SPACES TO KNOW

- 1 die: n = 6 · 2 dice: n = **36** (ordered pairs)
- n coins / n children: n(S) = $\mathbf{2^n}$ (2 → 4, 3 → 8, 4 → 16)
- 3 children: {BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG}
- Coin + die: 2 × 6 = 12 (H1…H6, T1…T6)
- Tools: **list** in a fixed order · **tree diagram** · **grid** (dice)

| SUM OF 2 DICE | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **ways /36** | 1 | 2 | 3 | 4 | 5 | **6** | 5 | 4 | 3 | 2 | 1 |

Doubles: 6 · sum odd: 18 · sum even: 18 · product odd: 9 (both odd)

## 4-1 · DECK OF 52 CARDS

4 suits × 13: ♥ ♦ red (26), ♣ ♠ black (26). Each rank (A, 2…10, J, Q, K) appears 4 times. **Face cards** J, Q, K = 12 (6 red). Ace is not a face card.

## 4-1 · THREE TYPES OF PROBABILITY

| | | |
|---|---|---|
| **Classical** | equally likely outcomes; count S | $P(E) = n(E)/n(S)$ |
| **Empirical** | observed data / frequency table | $P(E) = f/n$ |
| **Subjective** | expert's educated guess or opinion | no formula |

Tell: count S → classical · "of 500 surveyed…" → empirical · "a doctor estimates…" → subjective.

كلاسيكي = عدّ نتائج متساوية الفرص. تجريبي = من بيانات مجموعة f/n. شخصي = تقدير خبير.

## 4-1 · PROBABILITY RULES

1. $0 \le P(E) \le 1$ (negative or > 1 means an error)
2. P(E) = 0 → E is **impossible** (a 9 on one die)
3. P(E) = 1 → E is **certain** (a number < 7 on one die)
4. The probabilities of all outcomes in S add to **1**

Scale: 0 impossible · 0.25 unlikely · 0.5 50–50 · 0.75 likely · 1 certain

## 4-1 · COMPLEMENT

$$P(\bar{E}) = 1 - P(E) \cdot P(E) + P(\bar{E}) = 1$$

$\bar{E}$ = everything in S that is **not** in E (stay inside S!). Use it for: **not · none · neither · at least one**.

$$P(\text{at least one}) = 1 - P(\text{none})$$

المتمّمة: كل ما في فضاء العينة وليس في الحدث. استخدمها مع: ليس، ولا واحد، واحد على الأقل.

## 4-1 · THE FIVE PHRASES (VALUES 3, 4, 5, 6, 7)

| PHRASE | VALUES | SYMBOL | COMPLEMENT |
|---|---|---|---|
| exactly 5 | 5 | $X = 5$ | $X \neq 5$ |
| less / fewer than 5 | 3, 4 | $X < 5$ | at least 5 |
| **at most** 5 | 3, 4, 5 | $X \le 5$ | more than 5 |
| more than 5 | 6, 7 | $X > 5$ | at most 5 |
| **at least** 5 | 5, 6, 7 | $X \ge 5$ | fewer than 5 |

كلمة **at** تعني أن الرقم داخل الحساب، وكلمة **than** تعني أنه خارجه. ومتمّمة «على الأقل k» هي «أقل من k».

## 4-1 · SET OPERATIONS (VENN)

- **A ∪ B** "A or B": in A, in B, or both (inclusive)
- **A ∩ B** "A and B": in both
- **$\bar{A}$ = $A^c$** "not A": in S but not in A
- Check: n(A ∪ B) + n((A ∪ B)ᶜ) = n(S)

## 4-2 · MUTUALLY EXCLUSIVE (DISJOINT)

A and B **cannot occur at the same time**: no common outcomes.

$$A \cap B = \varnothing \Leftrightarrow P(A \cap B) = 0$$

Write the intersection, then decide. Die: even {2,4,6} & odd {1,3,5} → ME. Even & {1,2,4} share {2,4} → **not** ME.

If $P(A) + P(B) > 1$, A and B **cannot** be ME.

المتنافيان لا يقعان معًا (لا تقاطع بينهما). اكتب التقاطع أولًا ثم احكم.

## 4-2 · ADDITION RULES ("OR")

**Rule 1** (ME):

$$P(A \cup B) = P(A) + P(B)$$

**Rule 2** (not ME, always safe):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Rearranged:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

**3 ME events**:

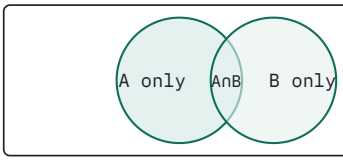
$$P(A \cup B \cup C) = P(A)+P(B)+P(C)$$

**3 events, general**:

$$P(A \cup B \cup C) = \Sigma P(\text{single}) - [P(A \cap B)+P(A \cap C)+P(B \cap C)] + P(A \cap B \cap C)$$

«أو» = جمع. إذا كان بينهما تداخل نطرح التقاطع مرة لأنه حُسب مرتين.

## 4-2 · VENN: THE FOUR REGIONS




| | |
|---|---|
| A only = A ∩ $B^c$ | P(A) – P(A∩B) |
| B only = $A^c$ ∩ B | P(B) – P(A∩B) |
| neither = $(A \cup B)^c$ | 1 – P(A ∪ B) |
| exactly one | P(A) + P(B) – 2P(A∩B) |
| $A^c$ ∪ B | 1 – P(A only) |
| P(A∩B) limit | ≤ smaller of P(A), P(B) |

Fill the overlap first, then the "only" parts; the four regions must add to 1.

## 4-2 · TWO-WAY TABLES

| | |
|---|---|
| P(A) | row/col total ÷ grand total |
| P(A and B) | **one cell** ÷ grand total |
| P(A or B) | (row + col – shared cell) ÷ total |
| P(neither A nor B) | 1 – P(A or B), or sum the other cells |
| ME? | the shared cell = 0? |

Two categories of the same variable (two colleges, two blood types) are always ME.

## TRAPS THAT COST MARKS

1. "or" read as exclusive: it means **or both**.
2. Rule 1 used when events overlap: double counts (king or heart = 16/52, not 17/52).
3. Multiplying instead of adding for ME events: P(A∪B) = P(A)+P(B), and P(A∩B) = 0.
4. Unequal outcomes treated as equally likely ({2 heads, 1 head, 0 heads} is wrong; use HH, HT, TH, TT).
5. 11 dice sums are **not** equally likely: always divide by 36.
6. "at least 2" ≠ "exactly 2"; "at most" includes the number.
7. $\bar{A}$ forgetting elements of S that lie in B.
8. Empirical: dividing by the wrong total; check Σf = n.

## WRITE IT LIKE A MARKER READS

1. Define events: "Let A = …".
2. Write S and E in **set notation** (the lecturer requires it).
3. Name the rule: "not ME, so Rule 2".
4. Substitute, simplify the fraction, give a decimal.
5. Check 0 ≤ P ≤ 1 and that complements add to 1.

عرّف الأحداث، اكتب فضاء العينة بصيغة المجموعات، سمِّ القاعدة، ثم عوّض، وتأكد أن الناتج بين 0 و 1.